

< JavaScript />

Activity06 – Variables_Operators_Arrays

You will work on three JavaScript files :

netId_Activity06_variables.js

netId_Activity06_operators.js

netId_Activity06_arrays.js

References to online tutorials and courses

<https://www.javascript.com/learn>

<https://www.w3schools.com/js/>

<https://developer.mozilla.org/en-US/>

<https://www.udemy.com/>

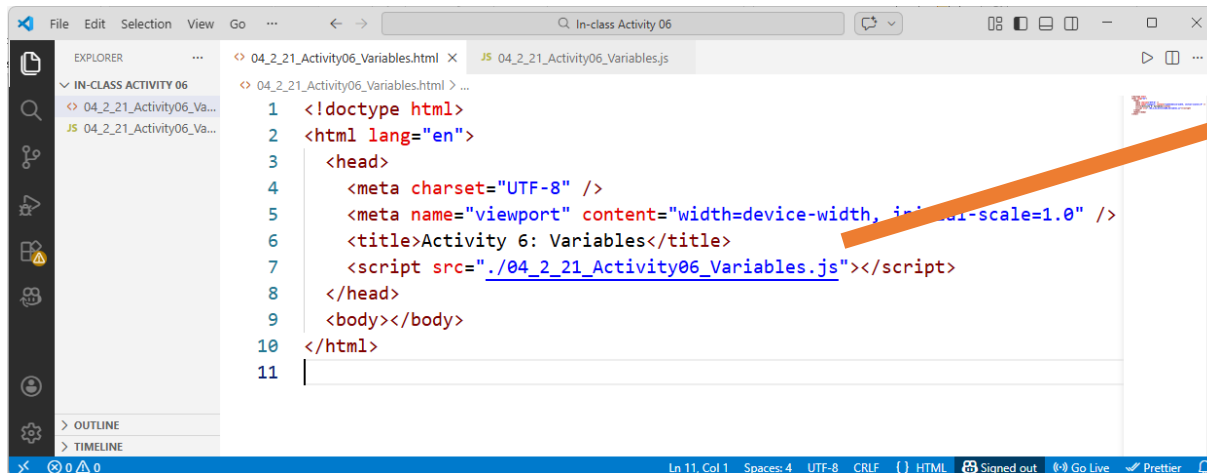


Write a HTML:

netId_Activity06_index.html

Write a JavaScript:

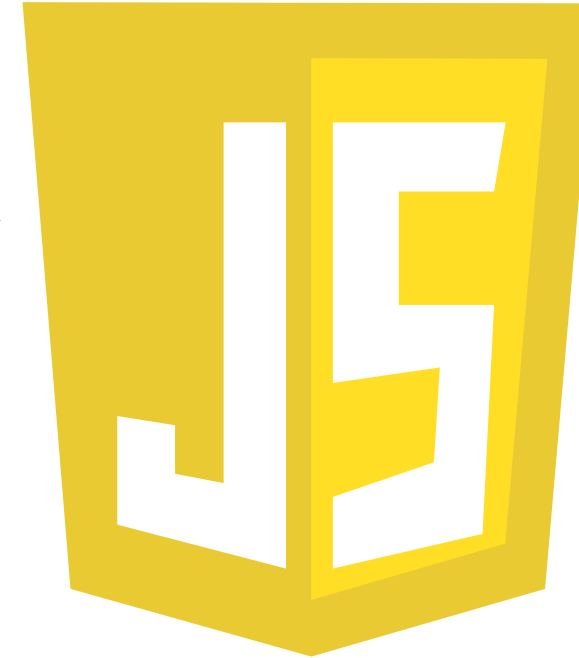
netId_Activity06_variables.js



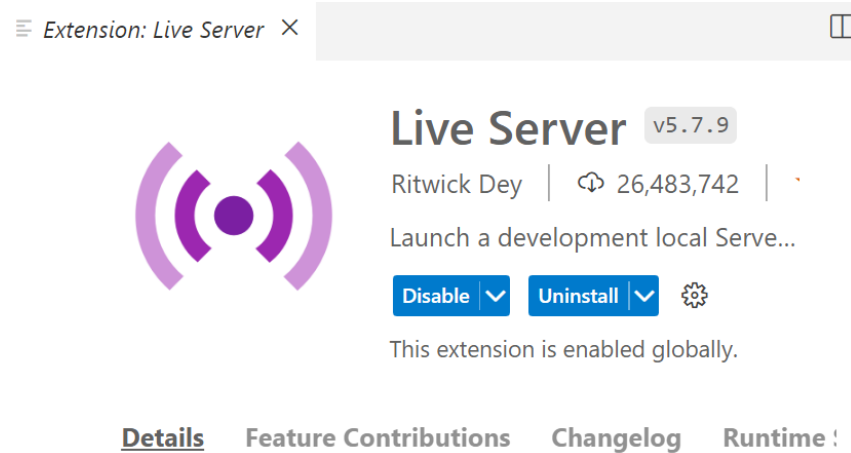
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a folder named 'IN-CLASS ACTIVITY 06' containing two files: '04_2_21_Activity06_Variables.html' and '04_2_21_Activity06_Variables.js'. The code editor shows the content of '04_2_21_Activity06_Variables.html' with the following code:

```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>Activity 6: Variables</title>
7     <script src="./04_2_21_Activity06_Variables.js"></script>
8   </head>
9   <body></body>
10 </html>
11
```

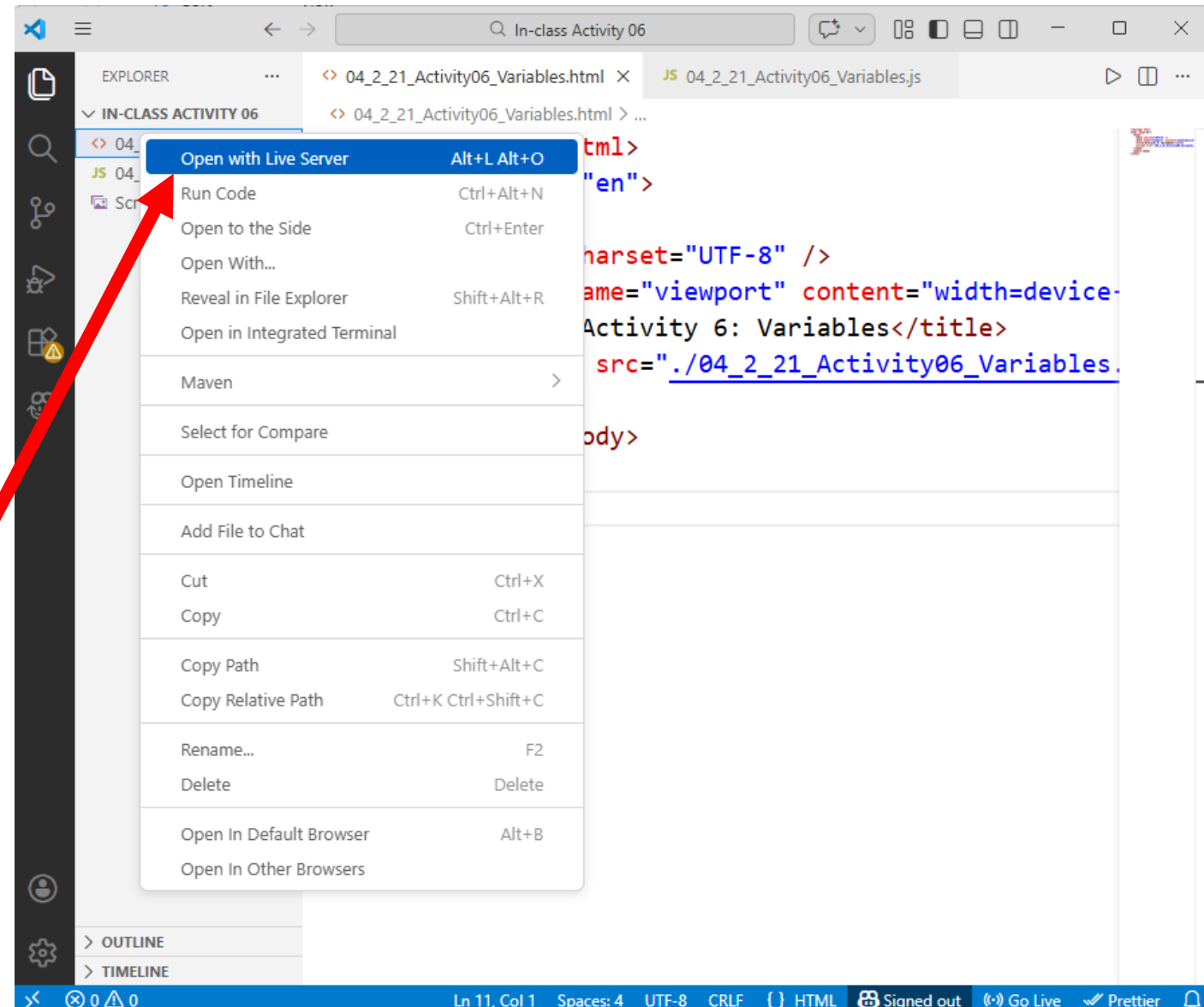
An orange arrow points from the script tag in the HTML file to the JavaScript logo on the right.



Run Live Server (Web server):



Live Server



Variables

In the JavaScript:

`netId_Activity06_variables.js`

Add your contact info and name of the Activity :

```
/*  
Name  
Feb 13, 2026  
Activity06 - Variables  
*/
```

```
console.log("---- I am in V A R I A B L E S ----")
```

Data types in JavaScript

```
graph TD; A[Data types in JavaScript] --> B[Primitive/ Value]; A --> C[Reference :]; B --> B1[String]; B --> B2[Number]; B --> B3[Boolean]; B --> B4[Undefined]; B --> B5[null]; C --> C1[Object]; C --> C2[Array]; C --> C3[Function];
```

Primitive/ Value

- String
- Number
- Boolean
- Undefined absence of a value
- null absence of an object

Reference :

- Object
- Array
- Function

Before 2015, there was only the keyword “**var**” in JavaScript to define variables.

2015 ES6 (ECMAScript 6) was created to standardize JavaScript.

And introduced **let** and **const**.

Let's see the difference between **var**, **let**, and **const**.

Write the next JavaScript :

```
// Q1 : Is it permitted the next ?
```

```
console.log("Q1 -----")  
var var1 = "Iowa";  
console.log(var1);
```



```
var var1 = 124;  
console.log(var1);
```

```
// Is it permitted ?  
console.log("Yes or Not");
```

```
// Q2 : Is it valid ?
```

```
console.log("Q2 -----");
```

```
let var2 = "Ames";
```

```
console.log(var2);
```



```
let var2 = 124;
```

```
// Is it valid ?
```

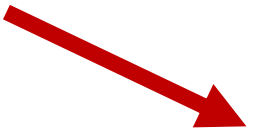
```
console.log("Yes or Not");
```

```
// Q3 : Is it valid ?
```

```
console.log("Q3 -----");
```

```
let var3 = "ISU";
```

```
console.log(var3);
```



```
var3 = 2023;
```

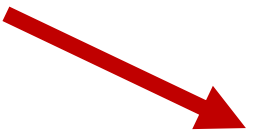
```
console.log(var3);
```

```
console.log("Valid ?")
```

// Q4 : Explain the Error.

```
console.log("Q4 -----");
```

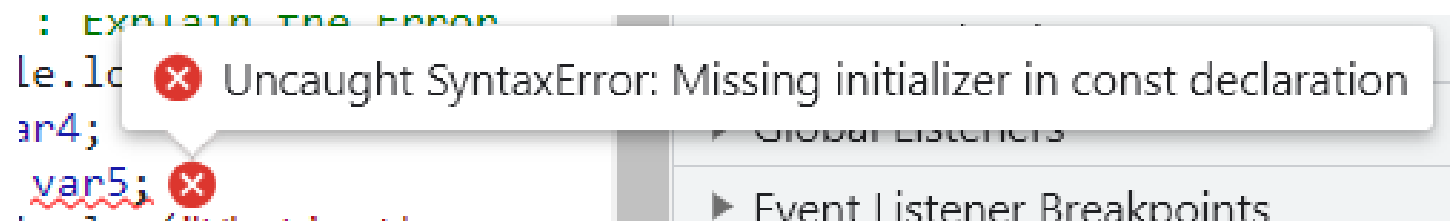
```
let var4;
```



```
const var5;
```

```
console.log("What's the error :")
```

Go to Browser>console and copy the error,
then answer the question.



// Q5 : Explain the Error.

```
console.log("Q5 -----");
```

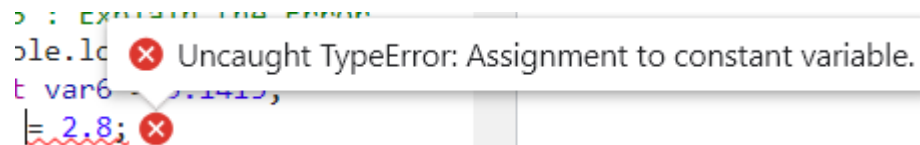
```
const var6 = 3.1415;
```



```
var6 = 2.8;
```

```
console.log("What's the error :")
```

Go to Browser>console and copy the error,
then answer the question.



```
// Q5 : Explain the Error.  
console.log("Q5 -----");  
const var6 = 3.1415;  
var6 = 2.8;  
console.log("What's the error :")
```

Uncaught TypeError: Assignment to constant variable.

// Q6 : Explain the Error.



```
let first name = "Sarfaraz";  
console.log(" ...replace this with your response.... ");
```



```
let 2numbers = [1,2];  
console.log(" ...replace this with your response.... ");
```



```
let city-state = "Ames Iowa";  
console.log(" ...replace this with your response.... ");
```

// Q7 : What !! ??

let mainCity = "DesMoines";

console.log("This is the Capital :", MainCity)



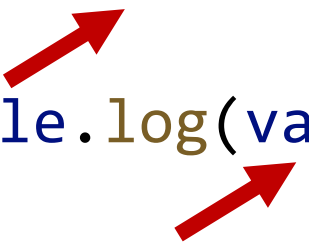
console.log("What's going on ?")

Go to Browser>console and copy the error,
then answer the question.

// Q8 : "let" and "const" scope vs "var" scope

```
if (5 === 5) {  
    var var7 = 100;  
}  
console.log(var7);
```

```
if (5 === 5) {  
    let var8 = 100;  
}  
console.log(var8);  
console.log("... explain ...")
```



Go to Browser>console and copy the error, then answer the question.

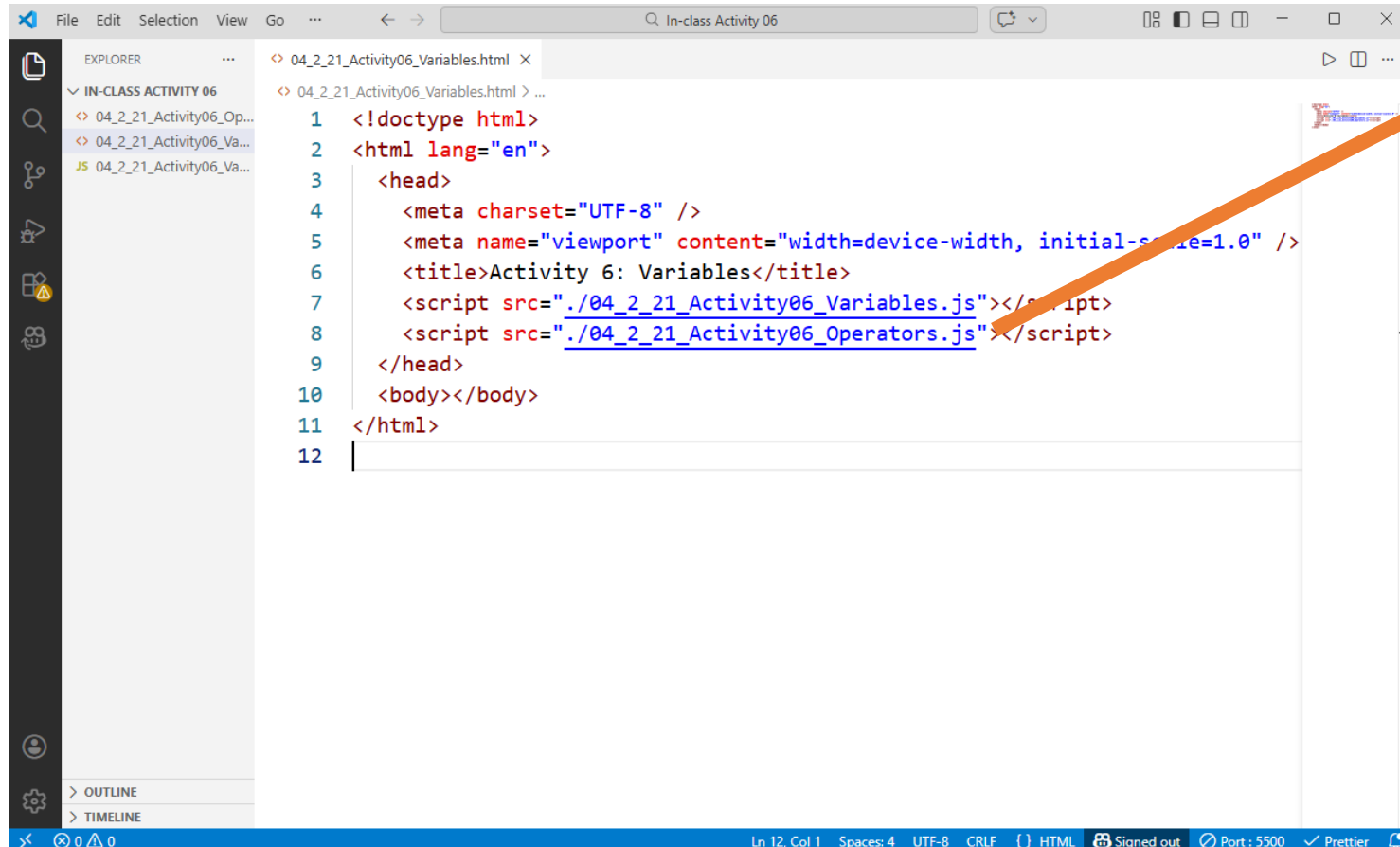
Operators

Add the JavaScript for Operators:

netId_Activity06_index.html

Write a JavaScript:

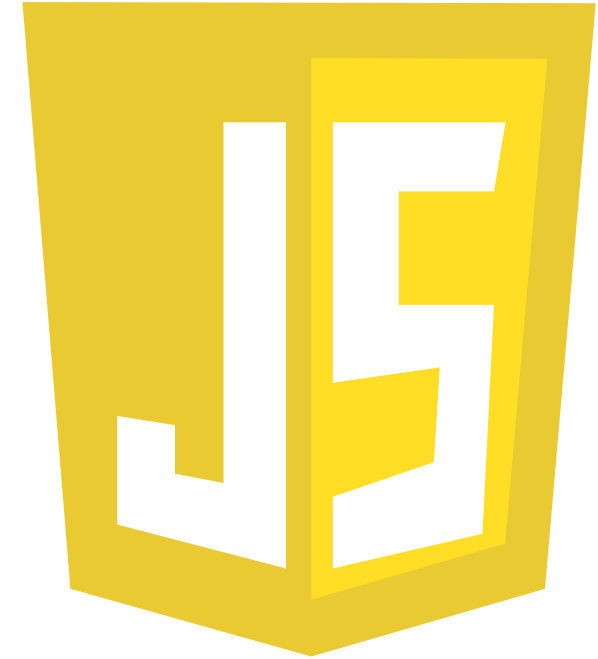
netId_Activity06_operators.js



The screenshot shows a web browser window with the address bar displaying "In-class Activity 06". The page content is an HTML document with the following code:

```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>Activity 6: Variables</title>
7     <script src="./04_2_21_Activity06_Variables.js"></script>
8     <script src="./04_2_21_Activity06_Operators.js"></script>
9   </head>
10  <body></body>
11 </html>
12
```

An orange arrow points from the second script tag in the code to the JavaScript logo on the right.



In the JavaScript:

`netId_Activity06_operators.js`

Add your contact info and name of the Activity :

```
/*  
Name  
Feb 13, 2026  
Activity06 - Operators  
*/
```

```
console.log("---- I am in O P E R A T O R S ----")
```

Priority | Precedence in Arithmetic, Comparison and Boolean operators

Priority	Operator	Description
1	()	Parenthesis
2	**	Exponentiation
3	+, -	Positive, Negative
4	*, /, %	Multiplication, Division, Modulus
5	+, -	Summation, Subtraction
6	<, <=, >, >=, !=, ==	Comparison operators
7	!	Boolean operator Not
8	& &	Boolean operator And
9		Boolean operator Or

Warning!

These are not summation, subtraction

Higher

Precedence

Lower

// Q1 : 5 + 9 / 2 + 1

Write a `console.log`
with the response of
the arithmetic
expression.

Test 9/2

// Q2 : 10 / 4 + 7 % 4

Test 10 / 4

Test 7 % 4

$$// \text{ Q3 : } (5 - 1) * ((7 + 1) / (3 - 1))$$

Observe the
precedence given by
parenthesis ()

// Q4 : $10 / 4 \geq 10 / 3$

What type of operator is $\geq$?

// Q5 : 10 >= 10 && 10 <= 10

What type of operator is & & ?

Equality Operators:

```
// Q6: Strict Equality  
// (Type AND Value)
```

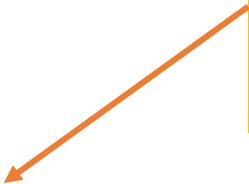
```
console.log(1 === 1)
```

true

```
console.log('1' === 1)
```

false

This is because the comparison between **string** and **integer**.



Equality Operators:

```
// Q7 : Loose Equality  
console.log('1' == 1)  
  
console.log(true == 1)
```

true

true

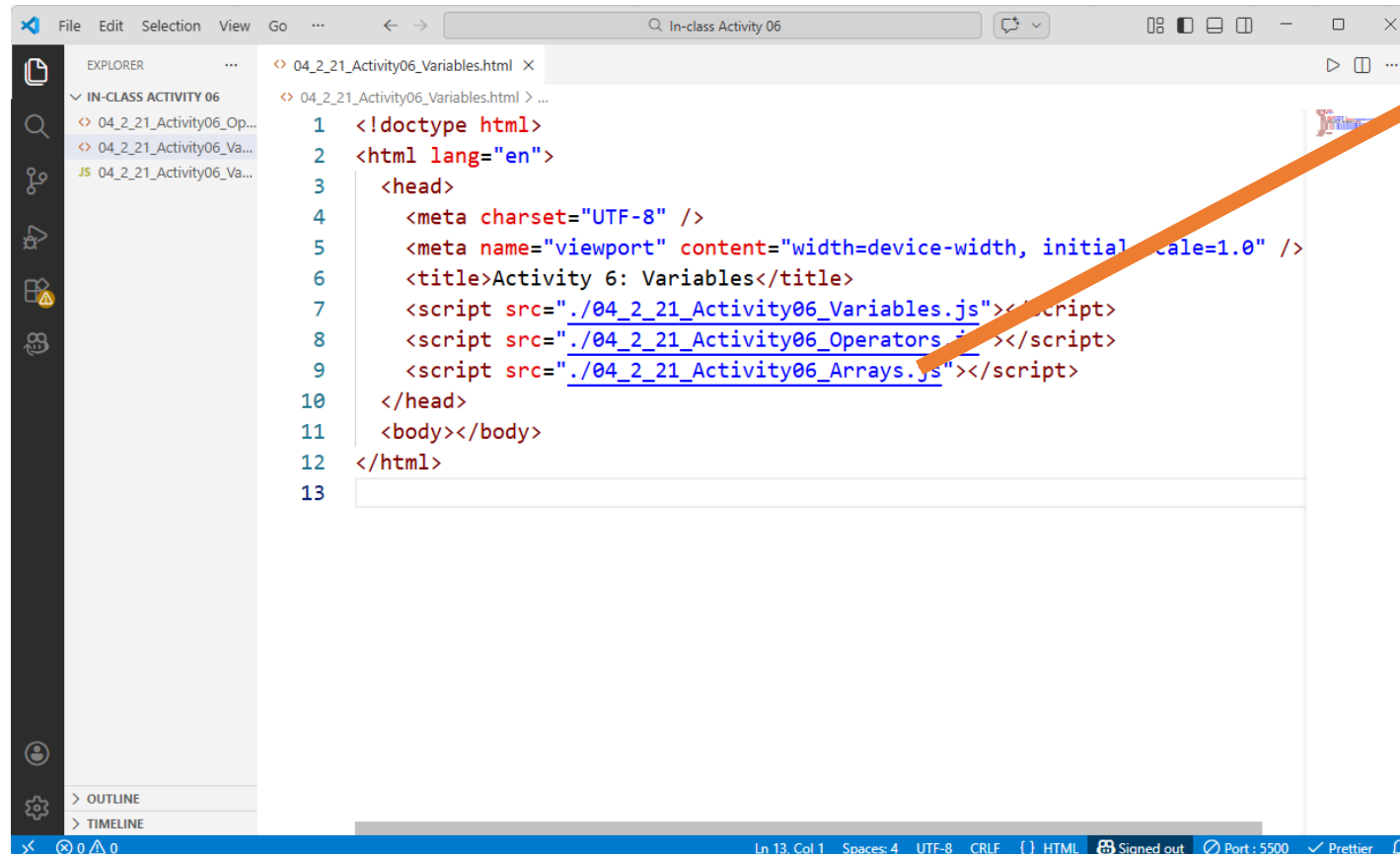
This is because the comparison between **string** and **integer**
It is true in loose equality.

boolean and **integer**
It is true in loose equality.

Arrays

Write a JavaScript:
`netId_Activity06_arrays.js`

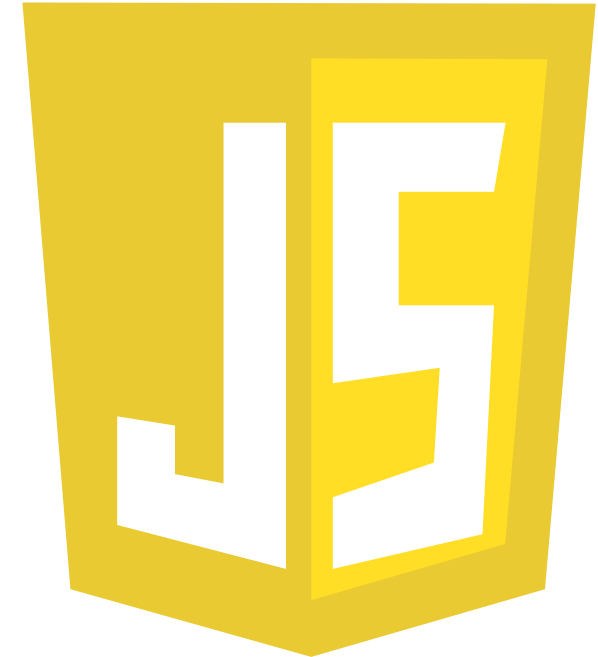
Add the JavaScript for Arrays:
`netId_Activity06_index.html`



The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a folder named 'IN-CLASS ACTIVITY 06' with three files: '04_2_21_Activity06_Op...', '04_2_21_Activity06_Va...', and '04_2_21_Activity06_Va...'. The code editor shows the content of '04_2_21_Activity06_Variables.html' with the following code:

```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6     <title>Activity 6: Variables</title>
7     <script src="./04_2_21_Activity06_Variables.js"></script>
8     <script src="./04_2_21_Activity06_Operators.js"></script>
9     <script src="./04_2_21_Activity06_Arrays.js"></script>
10  </head>
11  <body></body>
12 </html>
13
```

An orange arrow points from the third script tag in the code editor to the JavaScript logo on the right.



In the JavaScript:

`netId_Activity06_arrays.js`

Add your contact info and name of the Activity :

```
/*  
Name  
Feb 13, 2026  
Activity06 - Arrays  
*/
```



```
console.log("---- I am in A R R A Y S ----")
```

// Q1 : Index of each color

```
let colors = ['red', 'blue', 'orange']
```

e.g., `console.log(colors[0]);`

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5500/04_2_21_Activity06_Variables.html'. The browser's developer tools are open, with the 'Console' tab selected. The console shows a series of log messages from a JavaScript file named '04_2_21_Activity06_Variables.js'. The messages include: '...replace this with your response....', 'This is the Capital : DesMoines', '....What's going on ?', '100', '... explain ...', '---- I am in O P E R A T O R S --', 'true', 'false', 'true', '---- I am in A R R A Y S ----', and 'red'. Below these logs, the command '> colors' has been entered in the console. The result of this command is displayed as '< ▶ (3) [\'red\', \'blue\', \'orange\']'. An orange arrow points from the text 'Go to the Browser > console :' in the green box to the '> colors' command in the console.

```
Activity 6: Variables
127.0.0.1:5500/04_2_21_Activity06_Variables.html
Elements Console Sources Network
top Filter Default levels No Issues
...replace this with your response.... 04_2_21_Activity06_Variables.js:41
...replace this with your response.... 04_2_21_Activity06_Variables.js:43
This is the Capital : DesMoines 04_2_21_Activity06_Variables.js:46
....What's going on ? .... 04_2_21_Activity06_Variables.js:47
100 04_2_21_Activity06_Variables.js:52
... explain ... 04_2_21_Activity06_Variables.js:57
---- I am in O P E R A T O R S -- 04_2_21_Activity06_Operators.js:1
--
true 04_2_21_Activity06_Operators.js:2
false 04_2_21_Activity06_Operators.js:3
true 04_2_21_Activity06_Operators.js:4
---- I am in A R R A Y S ---- 04_2_21_Activity06_Arrays.js:1
red 04_2_21_Activity06_Arrays.js:3
> colors
< ▶ (3) [\'red\', \'blue\', \'orange\']
> |
```

Go to the Browser > console :

Type `colors`, the variable is in memory. The Browser recognize the value `colors`.

// Q2 : Add a new element 'green'

```
colors[3] = 'green';  
console.log(colors);
```

Do you remember
`colors.push()`?

What it means
`colors[3]` ?

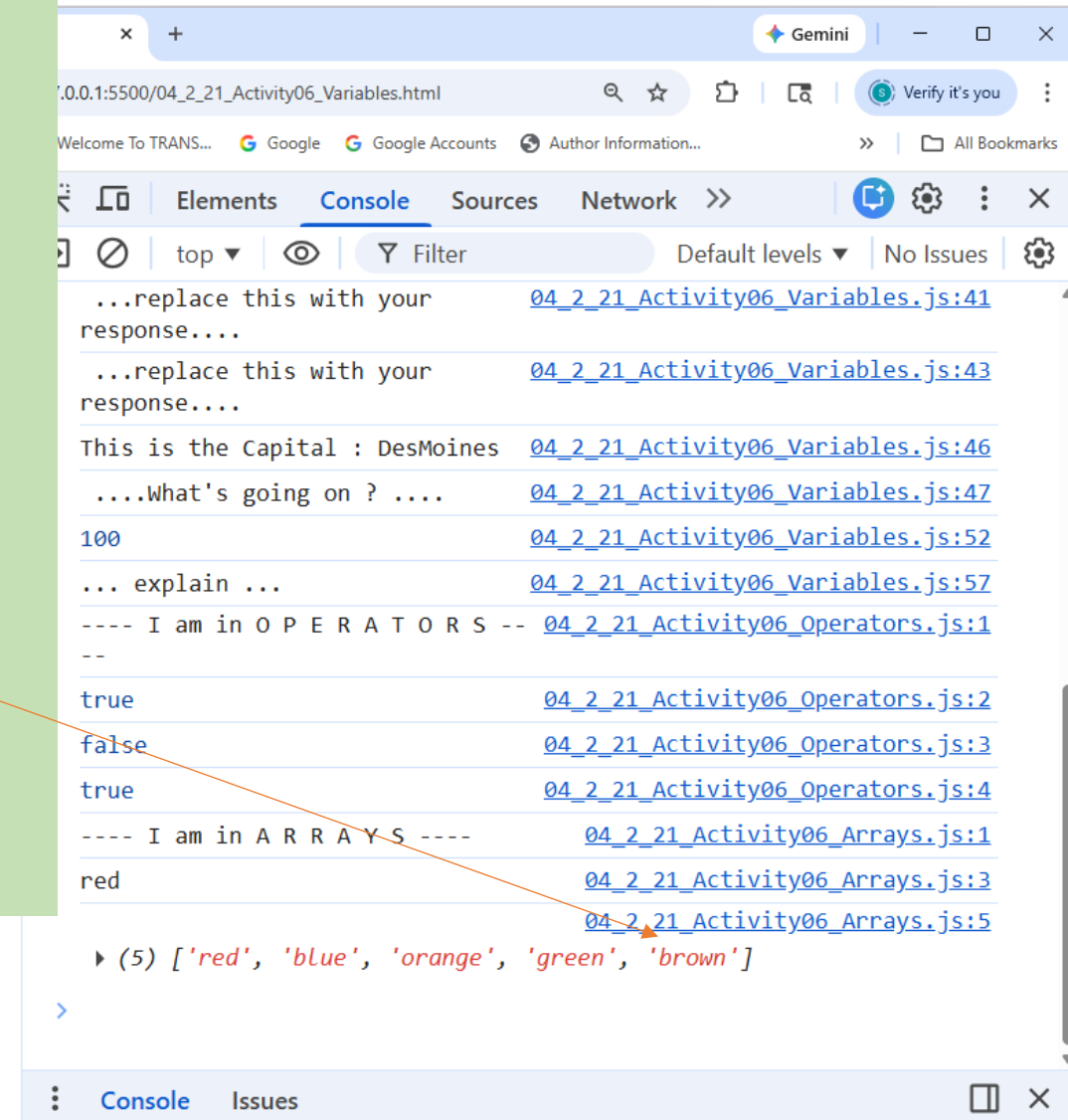
Test the next :

1. In the Activity6_arrays.js add a new color :

```
colors[4]='Brown';
```

2. Do not write `console.log(colors)` in the JavaScript file.

3. Goto to Browser, then console :
write `colors`.



// Q3 : Length of the Array

```
console.log(colors.length);
```

In VSC write `console.log(colors. ) ;`

Type
Ctrl-space